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1C - TNK

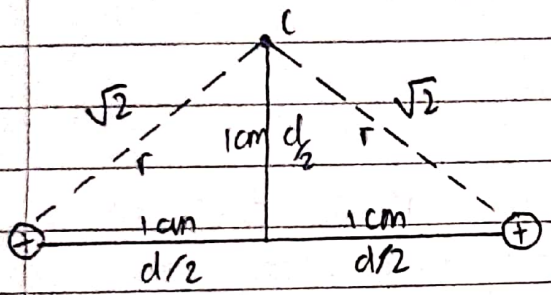
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1. Two charges $q = +2{,}0\ \mu C$ are fixed a distance $d = 2{,}0$ cm apart.

a) With $V = 0$ at infinity. What is the electric potential at point C?

b) You bring a third charge $q = +2{,}0\ \mu C$ from infinity to C. How much work must you do?

c) What is the potential energy U of the three charge configuration when the third charge is in place?



$r = \sqrt{1^2+1^2}$

$= \sqrt{2}$

a) Dik: $d = 2$ cm ; $q = 2\ \mu C = 2 \times 10^{-6}$ C

Dit : V ?

Jawab:

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r} + \frac{q_2}{r}\right)$$

$$= 9 \times 10^9 \left(\frac{2\times10^{-6}}{\sqrt{2}\times10^{-2}} + \frac{2\times10^{-6}}{\sqrt{2}\times10^{-2}}\right)$$

$$= 9 \times 10^9 \ (1{,}41 \times 10^{-4} + 1{,}41 \times 10^{-4})$$

$$= 9 \times 10^9 \ (2{,}82 \times 10^{-4})$$

$$= 2{,}538 \times 10^6 \text{ V}$$

b) $W = q.V$

$= (2 \times 10^{-6}) \times (2{,}538 \times 10^6)$

$= 5{,}076$ J

c) P.E system $= \frac{k.q_1q_2}{r} + \frac{k.q_2q_3}{r} + \frac{k.q_1q_3}{r}$

$= \frac{kq^2}{r} + \frac{kq^2}{r} + \frac{kq^2}{r}$

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$$= k\left(\frac{q^2}{r} + \frac{q^2}{r} + \frac{q^2}{r}\right)$$

$$= 9\times10^9\left(\frac{(2\times10^{-6})^2}{2\times10^{-2}} + \frac{(2\times10^{-6})^2}{\sqrt{2}\times10^{-2}} + \frac{(2\times10^{-6})^2}{\sqrt{2}\times10^{-2}}\right)$$

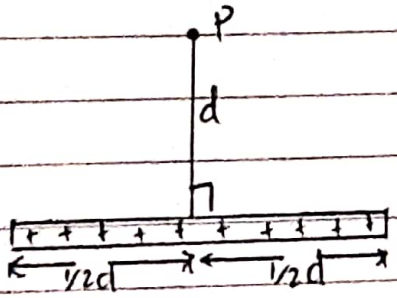
$$= 9\times10^9\,(2\times10^{-10} + 2{,}82\times10^{-10} + 2{,}82\times10^{-10})$$

$$= 9\times10^9\,(7{,}64\times10^{-10})$$

$$= 68{,}76\times10^{-1}$$

$$= 6{,}876\ \text{J}$$

2. Temukan potensial listrik di titik P dari distribusi muatan berbentuk batang seperti pada gambar rapat muatan panjang konstan λ



$$V = \int dv$$

$$= \int_0^2 \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{(x^2+d^2)^{\frac{1}{2}}}\,dx$$

$$= \frac{\lambda}{4\pi\varepsilon_0}\int_0^2 \frac{dx}{(x^2+d^2)^{\frac{1}{2}}}$$

$$= \frac{\lambda}{4\pi\varepsilon_0}\left[\ln\left(x+(x^2+d)^{\frac{1}{2}}\right)\right]_0^2$$

$$= \frac{\lambda}{4\pi\varepsilon_0}\left[\ln\left[L+(L^2+d^2)^{\frac{1}{2}}\right] - \ln d\right]$$

$$= \ln A - \ln B = \ln(A/B)$$

$$V = \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\frac{L+(L^2+d^2)^{\frac{1}{2}}}{d}\right]$$

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karena titik P berada tepat di tengah batang, maka jarak dari tengah menuju ujung batang sama antara arah kanan dan kiri, jadi persamaan tersebut dikali 2.

$$\therefore V = 2\frac{\lambda}{4\pi\varepsilon_0} \ln\left[\frac{L + (L^2+d^2)^{\frac{1}{2}}}{d}\right]$$

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